

Shivam Mittal

☎ (+1) 559-213-5557 | ✉ smittal39@wisc.edu | [in LinkedIn](#) | [GitHub](#) | [Portfolio](#) | 📍 Madison, WI

EDUCATION

University of Wisconsin–Madison

Aug. 2025 – May 2027

Master of Science (M.S.) in Computer Science, Research, Grade – 4.0/4.0

Madison, WI, USA

- **Coursework:** Natural Language Processing, Machine Learning, Foundation Models, Computer Vision, Distributed Systems

M.S. Ramaiah Institute of Technology

Aug. 2016 – Aug. 2020

Bachelor of Engineering (B.E.) in Electrical and Electronics Engineering, CGPA – 8.79/10

Bengaluru, India

WORK EXPERIENCE

Graduate Head Teaching Assistant – Computer Science, Part-time

Aug. 2025 – Present

University of Wisconsin–Madison

Madison, WI, USA

- Assist in teaching CS540 – Introduction to Artificial Intelligence, focusing on Deep Learning, Natural Language Processing (NLP), and Probability
- Create homework assignments, hold office hours, support students with concepts related to Machine Learning and AI

ZS Associates

June 2024 – Aug. 2025

Senior AI Software Engineer

Bengaluru, India

- Designed and deployed production-ready Generative AI multimodal agents with advanced text and vision capabilities leveraging Agentic Retrieval-Augmented Generation (RAG) improving response accuracy by **30%**
- Built GraphRAG, a knowledge graph algorithm for document QA, achieving **20% higher accuracy** than RAG agents
- Led multimodal, scalable Multi-Agent AI SaaS applications deployed across **3+ industries** with thousands of users
- Specialized in deploying local LLMs with sub-second latency, **100% data privacy** compliance for client-facing projects

Iron Mountain

Oct. 2022 – May 2024

Senior AI Software Engineer

Bengaluru, India

- Developed production-ready Intelligent Document Processing AI solutions achieving **99% data extraction accuracy** across diverse industries using Natural Language Processing and Computer Vision
- Engineered synthetic signature generation and developed a generalized signature detection model with **98% accuracy**
- Fine-tuned large language and multimodal models on domain-specific documents, improving performance by **15%**
- Developed production-ready AI SaaS applications for document question answering, deployed to **10+ clients**

TensorIoT

Oct. 2021 – Oct. 2022

Machine Learning Software Engineer

Bengaluru, India

- Developed time series forecasting models achieving **97% accuracy** to predict employee attrition rates
- Created a customized Alexa skill via the Alexa Developer Console to enhance customer purchase journeys
- Developed a deep neural network, achieving **98% accuracy** in detecting 15 distinct wall paint defects
- Engineered an end-to-end A/B testing MLOps CI/CD pipeline on AWS, adaptable for machine learning use cases
- Developed a customer journey tracking solution leveraging computer vision algorithms

ABB

Jan. 2021 – Sept. 2021

Computer Vision and AR/VR Researcher

Bengaluru, India

- Developed AI vision inspection for robotics manufacturing lines, boosting defect detection and reducing manual labor

Vijna Labs

July 2018 – Aug. 2018

Artificial Intelligence Intern

Bengaluru, India

- Worked on real-time object detection, image processing, computer vision, and training neural networks

PROJECTS & ACHIEVEMENTS

Research Projects – University of Wisconsin–Madison | *Madison, WI*

- [SparseZip](#) – Text-Aware Visual Token Selection and Compression for Efficient Vision Language Model Inference | [Report](#) | [Blog](#)
- [Market Prediction](#) with Vision Language Model using LLM post training, Reinforcement Learning and Agentic AI
- [Single Photon Challenge Competition](#) – Image reconstruction from single-photon noisy camera frames into a single high-quality image using RNNs, Stable Diffusion, VAEs, and GANs
- [Diffusion Language Models](#) – Project to investigate the practical differences between diffusion-based and auto-regressive foundation models and improvise generative capabilities of diffusion LLMs

U.S. Patent – Generative AI Agent for Intelligent Document Processing | *Filed, Pending with USPTO* | *Inventor*

- Developed a Generative AI Agent for intelligent document processing with a use-case driven routing mechanism

Undergraduate Projects – M.S. Ramaiah Institute of Technology | *Bengaluru, India*

- Built a self-driving electric vehicle using Deep Learning and NVIDIA Jetson Nano with **90% accuracy** on lane detection
- Led a team to design and develop a Formula Electric Vehicle. Achieved **first place** in Formula racing competitions

Awards & Hackathon

- **Awards:** Top Performer of the Year, Project Champion, InGenius Awards, and Employee of the Month
- **Hackathon:** Winner of the Iron Mountain AI hackathon, led a team of 3 members, awarded best engineering design
- **Hackathon:** Built [SellinSeconds](#), a university marketplace enabling students to buy and sell items in a few clicks, with a built-in peer transportation feature for seamless delivery assistance

TECHNICAL SKILLS

Core CS: Data Structures, Algorithms, System Design, Distributed Systems, Microservices, OOPs, High Performance Computing

Languages & Frameworks: Python, C++, PyTorch, Tensorflow, MySQL, Cypher, API Design, Git, FastAPI, ApacheAGE, GPU Programming

AI/ML: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Reinforcement Learning, OpenCV, Diffusion models

Generative & Agentic AI: Foundation Models, Vision-Language Models, LangChain, LangGraph, HuggingFace, Transformers, LLM Finetuning, Efficient model inference, Agentic Orchestration, Retrieval Augmented Generation, LLM Tool Utilization, Multi-turn Conversationality, LLMOps

Deployment & MLOps: AWS Services, AWS SageMaker, GCP, Azure, Docker, Kubernetes, GitLab, vLLM, Langfuse, LiteLLM